ARGUS: measured data and reference control

Licensed measured datasets, held-out evaluations, failure evidence and experimental nuisance-aware reference control.

Read this first

Measured tap screening has a useful plate-held-out result. Native ultrasonic localization and cross-panel quality rejection remain inadequate. Healthy-reference transport improves development waveform agreement, but separate confirmation shows worse adaptive localization and missed damage conditions. Zero false alarms does not rescue an insensitive detector. Development and confirmation remain distinct; the user apparatus, field coverage and patent novelty are unvalidated.

Contents

ARGUS measured-data campaign and results

Fundamental upgrade: reference-conditioned acquisition and decisions

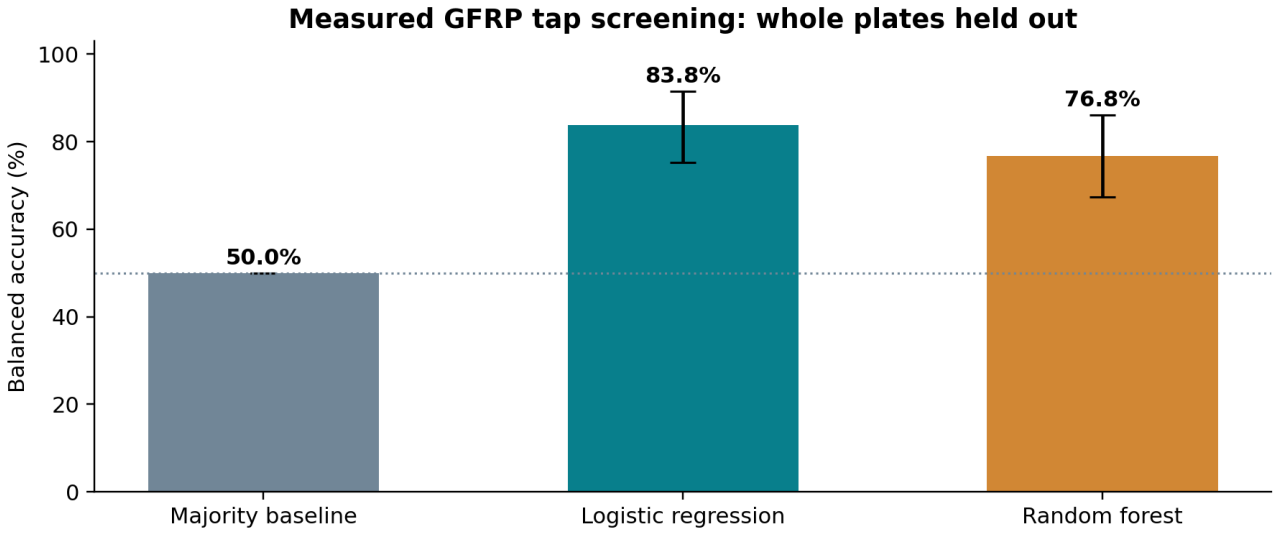
Reference-conditioned guided-wave fundamentals experiment

Healthy-only temperature and source-clock transport

Whole-trajectory reference gate for native guided-wave replay

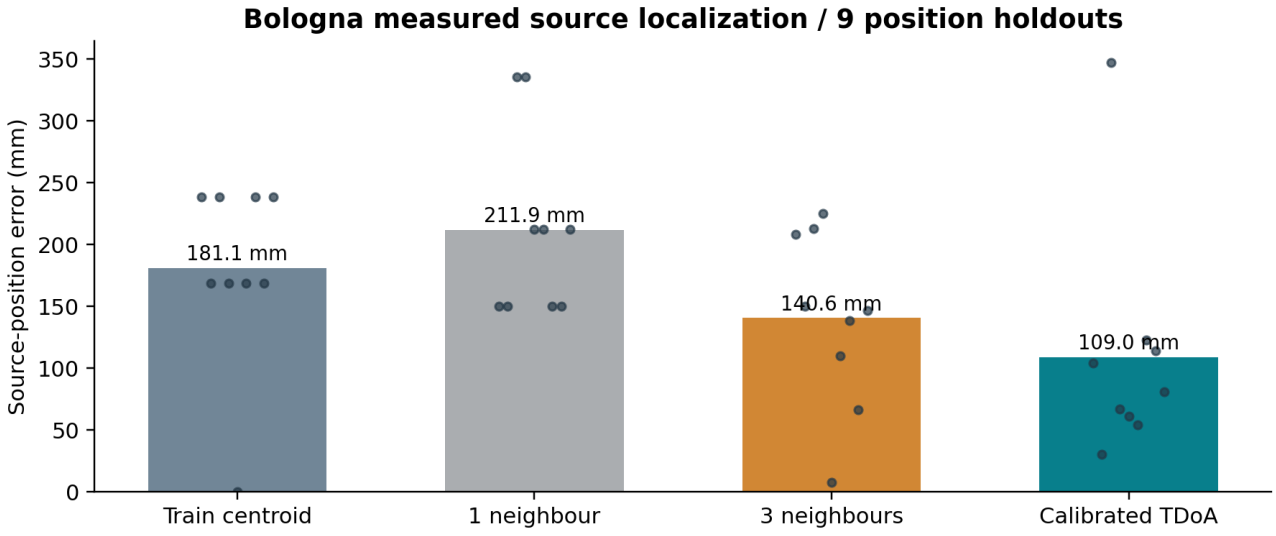
Source documents and exact commands remain in docs/. Current JSON and JUnit evidence remain in output/release/ and research_results/. The separate manuscript PDF is compiled from paper/main.tex.

Measured results at their actual scope



2,568 recordings / 8 plate clusters; bootstrap intervals condition on fitted out-of-fold predictions.

Figure 1. Pointwise audio-tap screening with entire plates held out. Eight clusters support the intervals.



Bars: mean; dots: retained position errors. One plate, previously seen data; no specimen-generalization interval.

Figure 2. Transient source-position error on nine previously seen records. Dots are retained position errors.

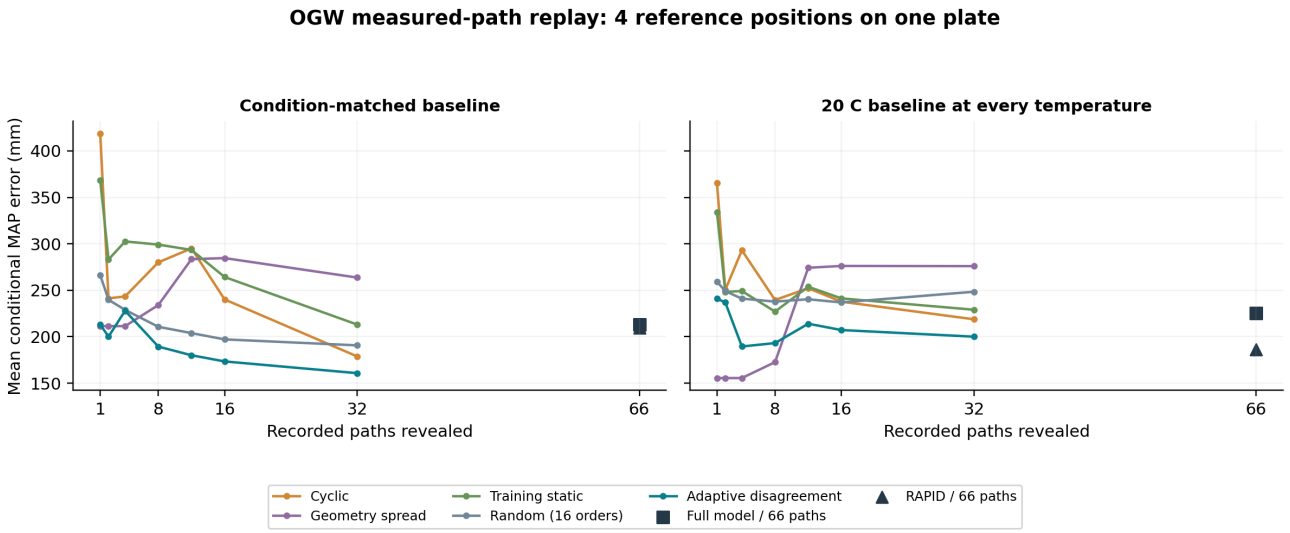


Figure 3. Conditional location error versus revealed paths. Full-bank references cost 66 paths; curves are not operational detection accuracy.

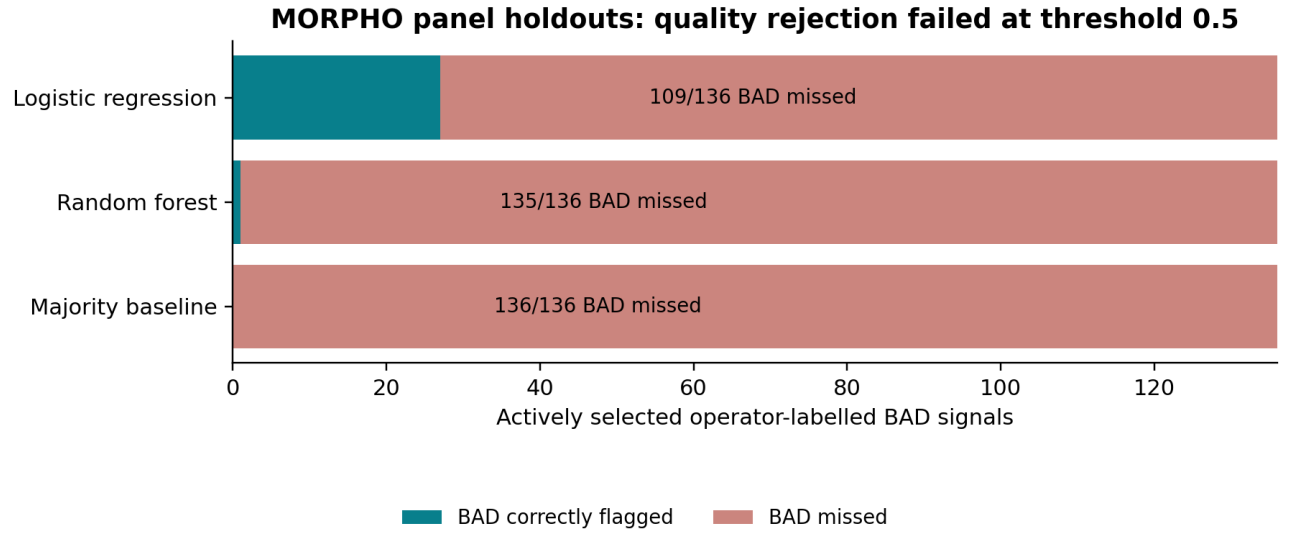
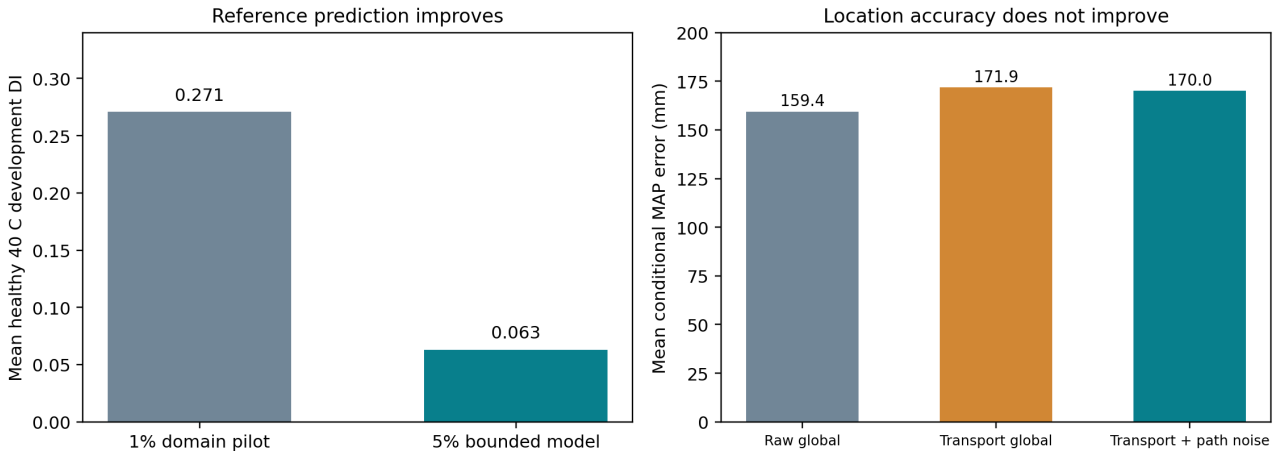


Figure 4. Cross-panel BAD-signal rejection fails despite apparently high overall accuracy.

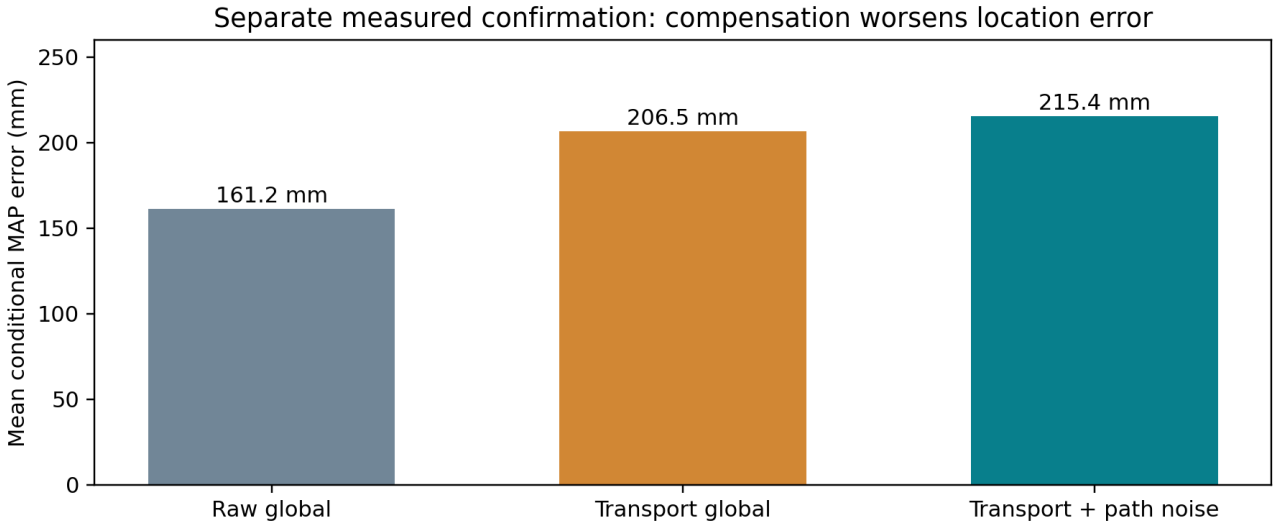
Reference control: measured development



Development only / one previously seen plate / shared 20 qualified paths / 16-path horizon; confirmation shown separately.

Figure 5. Healthy-reference development improves waveform agreement while conditional localization worsens. Only 20 of 66 paths qualify; the horizon is 16. These results reuse previously seen data.

Separate confirmation: negative result



Four mass-surrogate positions / two new test temperatures / one specimen; adaptive gate missed all eight damage conditions.

Figure 6. All 29 new banks were verified and scored with frozen parameters. Adaptive transported localization is worse than raw localization. All adaptive variants miss the eight damage conditions; zero false alarms on two unique healthy banks is not successful detection. Correlated calibration and one specimen leave field coverage unestablished.

ARGUS measured-data campaign and results

Prepared: 8 October 2026. Public measured-data evaluations, preserved with complete source and raw-data provenance.

What has now been achieved

Two new measured datasets were acquired: 18 complete Open Guided Waves banks (250.18 MB) and the MORPHO curated dataset (97.89 MB). The existing GFRP and Bologna recordings were also audited and reanalysed. No downloaded code or pickle model was executed. Public lab measurements were used directly at their documented sampling rates; missing paths, labels and reciprocal responses were not synthesized.

The strongest positive finding is plate-held-out audio-tap screening. Measured source localization also improves over a train-centroid baseline. The native ultrasonic path study exposes substantial location errors and healthy false positives. The cross-panel signal-quality classifier fails at its frozen threshold. These four outcomes describe different tasks and cannot be combined into one inspection-accuracy percentage.

Sources, licenses and actual independent units

Dataset	Measured subset and task	Independent units / license
TU Darmstadt GFRP	2,568 microphone taps, 51.2 kHz; pointwise intact/defect screening	Eight plates; CC BY 4.0
Bologna AE	Nine 2 MHz, three-sensor position records; transient source localization	One plate / nine positions; CC BY 4.0
Open Guided Waves 2	18 native 10 MHz banks / 40 kHz excitation / 66 actual paths	One plate / four removable mass positions; CC0
MORPHO	12,504 native 1 MHz signals; 1,011 selected GOOD/BAD labels	Five panels; labels actively selected; CC BY 4.0

OGW whole archives exceed 160 GB; bounded ZIP64 byte ranges retrieved only the selected full HDF5 banks. Each selected member has checked CRC32 and local SHA-256. The publisher whole-archive MD5 remains explicitly unverified. MORPHO files and the existing GFRP/Bologna source files were checked against published checksums. Exact URLs, actual temperatures, coordinates, schemas and licenses remain in acquisition manifests.

GFRP: a positive measured screening result

Four folds held out whole plates; all channels, points and repeats from a test plate stayed out of training. Features were recomputed from raw WAVs, filename/coordinate labels were excluded, scaling was fitted on training data, and all out-of-fold predictions were retained.

Model	Balanced accuracy	ROC-AUC	F1	Overall accuracy
Majority baseline	0.500	0.500	0.000	0.818
Logistic regression	0.838	0.922	0.649	0.833
Random forest	0.768	0.913	0.626	0.866

Logistic regression balanced accuracy is 83.8%, with a plate-cluster percentile interval of 75.2-91.5%. Its confusion counts are 396 detected defect-labelled taps, 72 missed taps, 1,743 correctly intact-labelled taps and 357 false defect-labelled taps. RF has higher overall accuracy but worse balanced accuracy, AUC and F1. Class imbalance makes the majority baseline 81.8% accurate despite detecting no defect-labelled taps.

The interval uses only eight plate clusters and conditions on fitted predictions; models are not refitted inside the bootstrap. Ten within-plate label permutation controls were near chance. This supports pointwise screening within this acquisition domain, not location, defect type, adaptive measurement efficiency or the user apparatus.

Bologna: measured source localization with strict position holdouts

Each complete position was held out. A bounded inverse-speed/two-relative-offset TDoA calibration used the other eight positions. The complete 1 m plate was searched on a fixed 5 mm grid. Three-neighbour preprocessing also used training data only. No numerical simulation data entered this reanalysis.

Method	Mean error (mm)	Median (mm)	Within 100 mm
Train centroid	181.07	168.75	11.1%
One neighbour	211.91	212.13	0.0%
Three neighbours	140.56	146.34	22.2%
Calibrated TDoA	108.99	80.78	55.6%

The 108.99 mm mean TDoA error is about 39.8% below the train-centroid error, but no held-out event is within 20 mm. This is a previously seen record on one plate, not new unseen validation or a persistent hidden-defect test. The CSV schema is 5,000 by three with columns s1/s2/s3; a separately identified repetition axis is absent even though the provider description mentions repeated experiments.

OGW: first native measured-path replay result

A new offline evaluation branch preserves 40 kHz/10 MHz observations instead of converting them into audio-range data. A fixed RAPID-style elliptical response model was calibrated on three reference positions; every temperature of the fourth position stayed held out. The second healthy thermal cycle was never used for calibration. Five policies revealed only the requested test path, with the same inference and path budgets.

The 960 trajectories include random-order repetitions and repeated trained models. They are not 960 independent physical trials: only four reference positions, twelve damage-condition banks and three unique healthy test banks support evaluation.

32-path policy	Matched mean conditional error (mm)	20 C baseline mean error (mm)	Matched conditional success within 50 mm
Cyclic	178.72	218.72	8.3%
Geometry spread	263.65	276.01	8.3%
Training static	213.04	229.04	8.3%
Random (16 orders)	190.70	248.45	8.9%
Adaptive disagreement	160.76	200.04	25.0%

Adaptive matched-baseline conditional MAP error is 160.76 mm versus random 190.70 mm. However, its model calls only 50% of the reference-scatterer conditions damaged and falsely flags 66.7% of healthy condition/model evaluations. Those evaluations reuse three healthy test banks under four trained models. Conditional distance success is not joint operational detection/localization success.

The full 66-path model has matched-baseline mean error 213.28 mm; classical full-bank RAPID has 209.43 mm. The full banks cost 66 reveals and are not equal-budget competitors. More evidence can worsen error when an observation model is misspecified. With a 20 C baseline at all test temperatures, adaptive mean error is 200.04 mm and there are no conditional 50 mm successes at 32 paths.

All trajectories completed with zero rejected opportunities. Modeled containment can be broad despite its apparent coverage; it is not a calibrated confidence region. The fixed global kernel/noise model and cross-date temperature/coupling differences remain major limitations. No beta, window or threshold was retuned on held-out results. A successful first run was preserved as a preflight pilot; the final rerun changed only validation/exception handling and meaningful causality tests. Every location metric remained identical.

MORPHO: a negative cross-panel quality result

Five complete panels were held out in turn. GOOD/BAD are actively selected human signal-quality labels, not defect labels. One panel has no BAD-labelled examples, so that fold has undefined AUC/balanced accuracy; it remains in the recorded evaluation. Actual HDF5 cycle counts disagree with the README, and the schema audit retains both.

Model	Balanced accuracy	BAD found / 136	BAD recall
Logistic regression	0.512	27 / 136	19.85%
Random forest	0.500	1 / 136	0.74%
Majority baseline	0.500	0 / 136	0.00%

RF misses 135 of 136 BAD signals at the frozen 0.5 threshold. Its 0.772 ranking AUC and 85.95% overall accuracy do not make it a working rejection gate. The majority baseline is 86.55% accurate while rejecting none of the BAD signals. Strong panel prevalence differences and selected labels limit interpretation. No threshold was retuned and no quality classifier was deployed.

What the project now contains

The repository contains reproducible public-data acquisition, native HDF5 validation, a causally restricted path oracle, whole-group evaluation, simple baselines, complete failures/negative findings, and source/raw hashes. The new replay module is for offline research and is separate from physical session execution. The operating apparatus still needs its own measured timing/transfer bench and blind physical inspection.

The full measured release verification passed 541 tests with one deliberate legacy-pickle skip. No physical probe was flashed or actuated for this campaign. Licenses and attribution remain with each source. Data acquisition alone does not establish patent novelty.

Reproduction

```
.venv\Scripts\python.exe scripts\download_open_guided_waves.py
.venv\Scripts\python.exe scripts\download_morpho_dataset.py
.venv\Scripts\python.exe scripts\validate_measured_gfrp.py
.venv\Scripts\python.exe scripts\validate_measured_impact.py
.venv\Scripts\python.exe scripts\validate_measured_guided_waves.py
.venv\Scripts\python.exe scripts\validate_measured_morpho_quality.py
.venv\Scripts\python.exe scripts\export_measured_web_summary.py
```

Preserve final reports. The impact/OGW commands refuse overwriting existing evidence; choose a new dated output when intentionally repeating them. The other commands should likewise be run with a new output path when retaining an earlier result. Dependency pins include h5py 3.16.0. All full study reports and protocols are under research_results/ and docs/. The Research page shows a compact task-separated summary.

Source-bound result artifacts

Study	Result file	SHA-256
GFRP	research_results/measured_gfrp_validation_2026_10_08.json	d6a4a536243bb928fc6e073775a6c00f771914b0a1bf27f4391e0e3d41da9edc
IMPACT	research_results/measured_impact_validation_2026_10_08.json	878533aff4d0df7623d76899ed745b569d81b075f86b888f130df0f35661dfe5
OGW	research_results/measured_guided_wave_validation_2026_10_08.json	4e3d97ca4cab63f97900363a76cbb74c65ba75624e58f014f4ccf03ae80cfbf6
MORPHO	research_results/measured_morpho_quality_2026_10_08.json	e034c9488f97d05ae2dbe78f5246e59c4c5659c57dec8ad47c0c1c7e9f4a22c1

Fundamental upgrade: reference-conditioned acquisition and decisions

Updated: 9 October 2026. Implemented and tested experimental code, with measured development and separate confirmation evidence; negative findings retained.

What changed and why

Real guided-wave recordings showed that temperature, source timing and waveform drift can resemble structural change. The initial model also applied an approximate spatial response as if its uncertainty were well described. The new chain treats reference validity, nuisance correction, spatial inference and anomaly decisions as separate operations that must agree before an unsupported signal can become an engineering claim.

1. Fit a healthy reference transport using only known healthy waveform anchors and actual surface temperatures. Record a per-path residual scale and reject non-identifiable or inaccurate fits.

2. For a requested path, align source timing from its captured electrical pitch. Predict the receiver reference before looking at the receiver response; a possible damage signal cannot fit away its own discrepancy.

3. Filter the feasible path bank by healthy-reference qualification. Use per-path observation uncertainty in the experimental adaptive disagreement criterion.

4. Compare accumulated discrepancy with complete healthy executions of that same bound controller. Use one maximum per whole trajectory, rather than pretending paths/checkpoints are independent calibration samples.

5. Withhold unsupported decisions and request qualified references. An anomaly requires a genuinely later, diverse verification with matching model/reference identities. All dispositions remain requests for review; none authorizes physical actuation or a healthy structural release.

Physics and mathematics at the supported scope

Temperature can change guided-wave timing, phase and amplitude; the dependence varies by material, mode and direction. The model approximates this effect with bounded affine time stretch, receiver delay and gain, conditioned on measured temperature. The recorded source pitch constrains source-clock drift independently of the receiver.

```
predicted_reference(t) = gain(T) * healthy_template(
    (t - observed_source_delay - receiver_delay(T)) / stretch(T))
corrected_DI = 1 - abs(correlation(predicted_reference, receiver))
path_variance = training_spatial_residual_variance + healthy_path_scale^2
adaptive_proxy = predicted_response_variance / path_variance
trajectory_score = max_over_fixed_checkpoints(mean(selected_DI^2 / healthy_scale^2))
reference_p = (1 + count(calibration_trajectory_max >= current_prefix_score)) /
  (N+1)
```

These are bounded local models and established statistical ideas, not newly discovered physical laws or a field-calibrated defect probability. Time stretching and phase compensation are discussed by Croxford et al. and Herdovics and Cegla. Reference ranks relate to established conformal prediction. The candidate technical subject is their specific ordered interaction with clock-described acquisition and reference-qualified control; novelty remains unestablished.

Healthy-only development result and qualification limits

The first 1% stretch-domain fit qualified zero of 66 paths. Its source, tests, protocol and failure report were archived exactly. The measured healthy development data showed that this domain was insufficient. A bounded 5% development revision retained every source-correlation, fit-error and identifiability threshold, and qualified 20 paths.

The mean leave-middle-temperature-out healthy discrepancy changed from 0.270896 to 0.063102, a 76.7% reduction. The diagnostic mean includes all 66 paths, including unqualified fits. It is not damage-localization accuracy. The independent healthy noise interpretation is also not established: the path scale is a development residual proxy.

Only 20 paths qualify across all three temperature anchors. Before opening any new confirmation results, the control horizon was fixed at 16, the largest declared checkpoint supported by that feasible set. All compared variants use the same 20-path bank; unsupported paths are not executed simply to fill a 32-path budget.

Actual measured development comparison

The experiment reanalyses previously seen OGW recordings. It holds out entire reference positions from spatial fitting, but it remains an exploratory development comparison on one plate. Three old healthy second-cycle banks serve only as partial development calibration; their later descriptive results are not independent healthy test performance.

16-path adaptive variant	Conditional location error (mm)	Gate outcome
Raw global response	159.45	Unsupported
Healthy transport / global noise	171.92	Unsupported
Healthy transport / path-specific noise	170.03	Unsupported

The compensated branches worsen localization relative to the raw response control. This negative finding is retained. Better healthy reference prediction does not establish that the spatial scatterer model is correct. The new compensated model remains experimental and is not enabled in production physical sessions.

The retained study contains 1,296 trajectories, including random-order repetitions and reused condition/model combinations. There is still one specimen and four reference positions, not that many independent physical trials. All source hashes remained unchanged during the final run. Controller identity/fresh-verification fixes were validated and rerun without changing predictive parameters.

The gate refuses a claim the data cannot support

At the fixed 5% level, 19 complete healthy blocks are needed even to obtain a minimum rank of 0.05. Three available development blocks give minimum rank 0.25. All current gate outputs therefore withhold supported anomaly decisions; reuse of a calibration artifact as test is separately rejected. Zero supported alerts here means insufficient evidence, not perfect detection or a healthy panel.

The maxima account for all declared checkpoint looks under an exchangeability assumption. Cooling and warming conditions are not proven exchangeable, and temperature blocks on one plate are dependent. No field 95% false-alarm certificate or specimen-generalization guarantee is claimed.

Separate confirmation has now been acquired and scored

The public source initially returned HTTP 403; those failed receipts and blocked scoring record remain preserved. Normal bounded public retries subsequently acquired all 29 original banks, totalling 403,070,304 bytes: 19 healthy cooling calibration blocks and ten distinct warming confirmation banks at nominal 30/50 C. All original selected-bank CRC32 and SHA-256 values were checked; the old 61 protected source/data files remained unchanged. No prior recordings, generated signals or model-weight files replaced the new captures.

The confirmation command checked exact roles, temperatures, cycles, original member identities, duplicates and original/development overlap before loading waveforms. The fitting references, qualified path set, coefficients, controller, horizon, scales, alpha and score were fixed before confirmation responses were opened. The run completed 864 trajectories and 216 controller/model calibrations, each reusing the same 19 physical healthy blocks; there were zero failed opportunities and all source hashes remained unchanged. These counts do not create 864 independent physical tests.

16-path adaptive variant	New conditional location error (mm)	Any-prefix supported damage alerts
Raw global response	161.20	0 / 8 damage conditions
Healthy transport / global noise	206.49	0 / 8 damage conditions
Healthy transport / path-specific noise	215.39	0 / 8 damage conditions

The separate confirmation repeats the negative localization finding. Each adaptive variant produced no supported alert in any of the eight mass/temperature conditions and no false alert on the two unique healthy confirmation banks. The healthy banks are evaluated under four held-position models. Zero healthy alerts alongside zero damage detections means the gate was insensitive, not a successful detector or a healthy-release certificate.

Seeded random orders with the two transported models yielded ten early alerts among 128 damage/order trajectories and none among 128 healthy/order trajectories; the raw random control produced none. These are repeated policy orders over the same eight damage conditions and two healthy banks, not ten independent physical detections. Final-budget-only reporting would miss those earlier alerts, so the retained report records any-prefix and earliest supported budget.

The calibration blocks are from one specimen and a cooling sweep; tests are warming conditions. Their exchangeability remains unverified. The completed acquisition does not establish a 5% field false-alarm guarantee, broader location accuracy or patent novelty. The compensated and reference-gated modes remain experimental, outside production physical acquisition. Better physical response modelling and independent specimen validation are still required.

Implementation, verification and reproduction

The current full backend and acquisition tests passed 662 cases with one deliberate legacy-pickle skip. Frontend tests, type checking, lint and production build also passed. No probe was flashed or physically actuated for these experiments.

Mechanism	Implementation
Immutable healthy-only warp / source alignment	backend/app/evaluation/healthy_wave_transport.py
Bound complete-trajectory calibration / coarse-rank refusal	backend/app/evaluation/trajectory_reference_gate.py
Request reference / fresh diverse verification / review	backend/app/control/reference_conditioned.py
Measured development or guarded confirmation	scripts/validate_reference_conditioned_replay.py
Separate acquisition preserving old measurements	scripts/download_ogw_confirmation.py

```
.venv\Scripts\python.exe scripts\download_ogw_confirmation.py
.venv\Scripts\python.exe scripts\fit_healthy_reference_development.py
.venv\Scripts\python.exe scripts\validate_reference_conditioned_replay.py --stage
  development
.venv\Scripts\python.exe scripts\validate_reference_conditioned_replay.py --stage
  confirmation
```

Existing evidence files are deliberately protected against overwrite. Preserve earlier outputs under a distinct dated name before intentionally repeating a study. The completed original confirmation receipt and all scored

trajectories are retained. The local Research page retains the measured-data task summaries; the new fundamental experiments are documented here and in their protocols, without automatic physical activation.

What this strengthens for technical and patent review

The implementation now makes the nuisance assumption testable, separates test response from reference fitting, limits actions to qualified references, calibrates the actual adaptive trajectory unit, binds control decisions to exact identities, and prevents stale verification or inadequate calibration from authorizing a claim. The disclosure retains the failures and acquisition gaps. These are concrete engineering improvements; patent novelty, inventive step, human inventorship and ownership still require review.

A suitable review statement is: "We measured where the model fails, improved healthy metrology and decision qualification, and tested fixed settings on separate measurements. Localization and damage sensitivity still failed, so those modes remain experimental. The next proof requires a better physical response model and a blind apparatus campaign."

Reference-conditioned guided-wave fundamentals experiment

8 October 2026. This protocol is fixed before the new confirmation waveform results. It follows a failed initial measured localization model and is an engineering development iteration, not a claim of an externally preregistered invention or unseen specimen generalization.

Technical change

The experimental chain separates source timing, healthy propagation drift and possible structural discrepancy. Healthy-only temperature anchors estimate a bounded receiver stretch/delay/gain model. The observed electrical source pitch aligns the clock of an acquired path. A test receiver waveform cannot fit its own reference warp. Per-path healthy residual uncertainty can temper spatial evidence and normalize the adaptive predictive-disagreement criterion. A separately calibrated complete-trajectory gate compares discrepancy to healthy executions of the same frozen controller before any anomaly is supported.

This directly addresses observed healthy false alarms and state-dependent acquisition. Baseline stretch, source correlation, heteroscedastic inference and conformal ranks are established methods; the candidate engineering subject is their ordered, reference-qualified interaction with acquisition and replanning. Novelty/inventive step remains a professional review question.

Data roles and blinding

Fit transport only from original healthy first-cycle warming banks at nominal 20/40/60 C, using actual surface readings and recorded source pitches. The healthy 40 C leave-middle-out development check estimates a residual proxy from 20/60 C interpolation, with fixed floor 0.02. Final fitting can use all three anchors. These fit references are not independent physical noise estimates.

The new acquisition plan reserves 19 healthy first-cycle cooling blocks (22..58 C, step 2) for trajectory calibration, and separate 30/50 C warming banks for confirmation: two healthy second-cycle banks and eight reference-mass banks. No confirmation receiver waveform can fit transport, reference scales, policy, alpha, score or threshold. A full comparison must wait for all 29 original members to arrive with verified provenance. Missing data is never synthesized.

If the repository cannot supply those banks, only an explicitly labelled development reanalysis of the original 18 banks may be run. The previous positions and results were already seen. Such a reanalysis is exploratory; it cannot be relabelled as fresh confirmation. The healthy-only fit remains a useful metrology-development result even without confirmation captures.

Prespecified model and controls

Keep the native 40 kHz/10 MHz recorded windows, 66 actual unordered pairs, 50 by 50 full-panel grid and beta=1.05 spatial kernel. A whole location is held out of the spatial fit, including all its development temperatures. The source pitch and temperature of a selected test path are observable acquisition data; unselected test pitches and receivers remain hidden from path selection.

Compare raw-global, transported-global and transported-heteroscedastic response models on a shared path set qualified by healthy-anchor support. Unsupported source/temperature observations consume their opportunity but cannot update location or pass the gate. Qualified-path shortages remain explicit failures.

For heteroscedastic inference, the squared residual mean of the training spatial model is added to the independently fitted healthy path scale squared, with floor 0.02. The adaptive criterion is predictive response variance divided by that declared observation variance. This is a disagreement proxy, not exact information gain. A learned spatial response is not a defect-existence model.

The first healthy-only fit qualified zero paths under a 1% stretch domain. Its exact failed source and report are preserved. A 5% physically bounded development revision keeps all fit/source/identifiability qualification thresholds and qualifies 20 paths. Before confirmation analysis, freeze the largest stated checkpoint below that number: horizon 16. Every variant uses those same 20 feasible paths, rather than executing unsupported references.

Use adaptive, geometry-spread and 16 fixed seeded-random policies with budgets 1/2/4/8/12/16 and seed 20261008. Keep every failed opportunity, conditional location error, source/transport rejection, class weight and query count. No parameter is selected by confirmation label accuracy.

Gate and control disposition

Calibrate one maximum checkpoint mean squared standardized discrepancy per complete healthy controller execution. Binding hashes join the exact model, controller, reference and protocol. A new prefix is compared conservatively using `(1 + calibration maxima >= prefix score) / (N+1)`. A rejected reference, model mismatch, reused artifact, incomplete trace or fewer than 19 blocks withholds the fixed 5% decision.

The required feedback is explicit: unsupported evidence requests qualified reference acquisition/recalibration; supported anomaly requests an independent verification; background-consistent evidence can continue within budget. There is no automatic healthy structural release or actuator command. The fixed maximum across complete trajectories accounts for all stated looks, subject to exchangeability assumptions. Cooling versus warming and one-specimen dependence leave those assumptions unverified; no field 95% certificate follows.

Evidence boundary

Report healthy reference prediction separately from conditional location and operational anomaly decisions. Gate refusal is not detection accuracy. Report calibration cost and count as whole blocks, not the number of paths. Preserve all old reports and source versions. Experimental modules remain separate from the production audio-range physical session until independently validated.

Healthy-only temperature and source-clock transport

Written 8 October 2026 before inspecting confirmation catch signals or damage indices. This is a new experimental native-frequency compensation module; existing studies, protocols and raw manifests remain unchanged.

Preserved healthy-development failure and revised domain

The initial one-percent stretch policy qualified **zero of 66 receiver paths** across all healthy anchors. The healthy source alignments remained supported, but receiver fits frequently reached the stretch bound or had excessive residuals. That failure, its report, and the exact initial module/tests/protocol are archived under `output/release/fundamental_pilots/transport_v1/`, with SHA256 receipts. The original report remains historical evidence.

Before any new confirmation catches are used, the healthy-development revision widens the candidate stretch domain to **±5%** and the fixed coarse grid to **41 by 41**. This is an explicitly documented development choice from healthy data, not a claim that all thermal propagation is bounded by five percent. Residual, identifiability, source, receiver-delay and gain qualification limits are unchanged. A wider search is not permission to declare a poor fit supported.

Every fitted model owns an immutable `FitPolicy`; prediction uses that model's exact bounds. The model identity binds the complete policy, training-array hash and implementation-source SHA256. No module-global monkeypatch changes a fitted model. An explicit one-percent policy remains available for reproduction. Later experiments must record which policy they use; the failed policy cannot be overwritten or silently replaced in an earlier result.

Established physical basis

Temperature changes elastic propagation and transducer response. Baseline signal stretch and optimal baseline selection are established methods; see Croxford, Moll, Wilcox and Michaels (2010). Combining propagation stretch with phase-response compensation also has prior literature, including Herdovics and Cegla's transducer phase-response work. This implementation makes no claim that those methods or an affine time warp are new inventions.

The approximation is `healthy_receiver_T(t) ~= gain(T) * receiver_20((t - source_delay - receiver_delay(T))/scale(T))`. A scale above one lengthens propagation time. Source delay is estimated from the **recorded electrical pitch**, separately from receiver propagation. The approximation cannot represent arbitrary dispersion, mode conversion, temperature gradients or coupling hysteresis.

Inputs and prohibition on test fitting

Use only the existing first healthy ascending-cycle anchors at nominal 20, 40 and 60 C, with their actual two-sensor surface temperatures. Inputs are three complete native catch/pitch banks, 13,108 samples per recorded path at 10 MHz. The measured study uses all 66 paths; synthetic contract tests may use fewer paths. No damage position, damage waveform, confirmation catch, second healthy cycle, new 30/50 C confirmation bank or 19 cooling calibration blocks may fit or tune transport parameters.

The source template and receiver template are the first healthy anchor. Immutable bytes-backed arrays prevent callers from modifying fitted templates or coefficients. The module has no dataset loader, RNG, truth argument, adaptive window selection or serialization of executable model objects.

Fixed fitting and parameter domain

Record source-clock delay using signed FFT cross-correlation in a fixed first-250-microsecond pitch window, restricted to ±10 microseconds. Use a three-point peak refinement. Reject zero-energy/nonfinite pitch, a bound-hitting peak or aligned signed correlation below 0.80. These limits are fixed before confirmation and narrower than half a 40 kHz cycle; no damaged receiver participates in this alignment.

For each non-reference healthy anchor/path, align out its measured pitch delay, then fit receiver stretch, residual delay and positive gain against the healthy reference. The revised default uses a fixed 41-by-41 coarse grid followed by bounded least-squares refinement with at most 60 function evaluations. The optimization uses every fourth native sample, still well above the excitation Nyquist rate; prediction and damage-index calculation retain the full native window.

The revised default bounds are scale 0.95-1.05, receiver delay ±10 microseconds and gain 0.5-2.0. Gain is profiled analytically for each warp. Outside-template times receive zero baseline prediction; their count is disclosed. A dimensionless finite-difference Jacobian of normalized healthy residuals supplies a scale/delay identifiability diagnostic, with a fixed condition-number limit of 10,000. A normalized healthy fit RMS above 0.30, active parameter bounds, or insufficient source alignment marks a fit unsupported. The limits are conservative model-policy choices, not instrument specifications. Explicit policies may tighten these qualification limits; they cannot relax them to force a successful development score.

Interpolate the three healthy parameter anchors piecewise linearly in the mean actual surface temperature. Permit at most 3 C extrapolation outside those means, disclose it, and reject larger excursions. A new pitch waveform may supply only the selected path's source-clock alignment. It must not be acquired or analysed across future unselected paths to rank them.

Independent development residual and downstream gate

Before the middle healthy anchor is incorporated into the final interpolation, predict it from the 20/60 C endpoint fits and its own recorded pitch. The **old damage index**, `1 - abs(Pearson(predicted_reference, observed_catch))`, remains unchanged and bounded in [0,1]. This middle-anchor prediction is a healthy-only development diagnostic.

Freeze each path's scale as `max(0.02, middle-anchor development damage index)`. This is a conservative residual normalization proxy from one held-out temperature, not an independently estimated noise standard deviation or calibrated probability. The 19 new complete healthy cooling blocks must not change it; those blocks are reserved for separately calibrating a complete adaptive trajectory gate. New confirmation catch signals remain untouched until all transport/gate source, tests and protocol are frozen.

The final model may use all three healthy anchors for interpolation. Its prediction API accepts only path index, actual surface temperature and that path's recorded pitch. Observed catch is accepted by a separate damage-index operation **after** the reference has been generated, and cannot change the warp or fitted model. Return the residual-normalization scale, corrected index, normalized RMS diagnostic, fit/source/temperature diagnostics and supported status. Unsupported observations consume the acquisition opportunity and remain rejected in downstream traces.

Verification and limits

Tests cover recovery of injected healthy delay/stretch/gain, electrical source-delay sign, fixed pitch bounds, immutable templates, invalid numeric/native shapes, temperature extrapolation, residual-scale separation, and the fact that changing a test catch cannot change the predicted reference. No test asserts superiority on a fabricated damage label. No confirmation outcome changes a parameter or window.

This module compensates nuisance variation using measured healthy development data. It does not validate the physical specimen, source-clock hardware, a defect mechanism, cross-specimen generalization, exchangeability of thermal cycles, or a field probability. Its actual measured effectiveness must be reported from the subsequently frozen confirmation study, including rejection or failure.

Whole-trajectory reference gate for native guided-wave replay

Date: 8 October 2026. Status: prespecified experimental method, implemented in `backend/app/evaluation/trajectory_reference_gate.py`. This document fixes the score and calibration procedure before confirmation damage-bank results are used. It does not report an executed physical experiment or a measured benefit. The production audio-range inference engine is not automatically changed.

Healthy-only feasibility revision before confirmation

The exact original horizon-32 gate, its 53 passing focused tests and this protocol's first version are preserved with SHA256 receipts under `output/release/fundamental_pilots/gate_horizon32/`. That version remains a historical source snapshot, not evidence that a 32-path measured reference trajectory was qualified.

Before any new confirmation bank was available or opened, the healthy development study found that a +/-1% receiver time-warp bound qualified zero paths. A versioned +/-5% physical time-warp bound, retaining the same waveform, Jacobian and source qualification thresholds, qualified 20 of 66 paths across the 20/40/60 C development references. The development leave-40-out corrected DI mean was approximately 0.0631 compared with 0.2709 for the stated baseline. These are healthy development results only; neither damage detection nor unseen-temperature confirmation is inferred from them.

The new experiment therefore freezes **horizon 16**, the largest existing prespecified checkpoint below 20 qualified paths, and the checkpoint prefix `(1, 2, 4, 8, 12, 16)`. The controller must operate within a healthy-development feasibility mask fixed before confirmation. The gate itself does not verify that mask; the model/controller/protocol hashes and experiment provenance must bind its actual content. An unsupported queried observation still makes the decision unsupported; it cannot be omitted to maintain the trajectory length.

The API now accepts only an explicitly declared nonempty **prefix of the original checkpoint tuple**. Full horizon 32 remains the API default for the original method; arbitrary budgets, skipping intermediate checkpoints, changing the declared horizon while testing, and adding later looks are rejected. A complete calibration trajectory reaches the horizon declared by its binding: 16 for the new feasibility-limited experiment, 32 for the original binding. Those bindings cannot share a calibration. The fitted model, controller and protocol hashes must include the exact horizon/checkpoints. Nothing about this revision changes the 19-whole-block minimum p=0.05 resolution, the separate qualification requirements, or the unverified cooling-versus-warming exchangeability. The common horizon below is understood as the binding's declared horizon; references to 32 describe the original default protocol.

The problem being addressed

A controller chooses later transducer paths from earlier responses. Those responses are therefore neither a random sample of all paths nor independent observations. Repeating a per-path anomaly test at each acquisition can also raise an alarm eventually just because the algorithm had many opportunities. Temperature can change a healthy waveform enough to resemble damage.

The experiment separates three operations: estimate a healthy response from fitting references; execute a frozen controller that reveals one recorded path at a time; compare its accumulated discrepancy with complete healthy executions of that same controller. The latter comparison calibrates the unit that will actually be monitored: an entire adaptive trajectory, rather than one channel.

Frozen inputs and calibration unit

The measured-data experiment uses the Open Guided Waves temperature extension: native 40 kHz, five-cycle excitation, 10 MHz sampling, 66 recorded pairs among twelve transducers on the same 500 x 500 x 2 mm CFRP plate. See the source audit and the authors' Scientific Data article.

The planned new calibration set is 19 first-cycle healthy **cooling** banks at nominal 22, 24, ..., 58 C. The condition used for reference transport is the recorded surface temperature, not a silently substituted chamber setpoint. Each native bank has 66 paths; the original default controller replay reaches 32 distinct acquisition opportunities, while the revised healthy-feasible experiment reaches 16. Checkpoints are the binding's declared prefix of `(1, 2, 4, 8, 12, 16, 32)`. A separate complete healthy trajectory is replayed for every model/controller binding. A bank contributes exactly **one** scalar maximum for each such calibration, not 66 independent samples or seven independent checkpoint samples.

These are 19 temperature-condition blocks from **one physical specimen**. They are not 19 independent plates, trials after independent remounting, or examples drawn randomly from a field operating population. Reusing the banks for another controller or fold does not create additional independent physical specimens.

The healthy reference transport and each path's scale are fitted using only the declared development healthy references. In the accompanying experiment, those are first-cycle warming banks at nominal 20, 40 and 60 C. The 40 C development reference may be left out temporarily to estimate residual scale against interpolation from 20/60 C, after which the declared final reference model can use all three. The 19 cooling calibration banks do **not** fit the reference transport, path scales, response model, policy, score or alpha. Confirmation warming conditions at 30/50 C are likewise excluded from fitting.

Download/access failures and unsupported acquisitions remain in the study accounting. Only actually acquired, complete and qualified healthy replays may be passed to the calibrator. The module refuses an invalid calibration batch; it does not discard malformed blocks to make an alarm threshold easier to meet. If fewer than 19 valid blocks are available, the smaller N and resulting resolution are reported, and the 5% gate returns `unsupported`.

Fixed causal score

At path acquisition j, the reference transport returns corrected damage index `d_j`, retaining the existing `1 - abs(Pearson correlation)` definition, and a finite positive scale `sigma_j` fitted from healthy development data. The gate also permits a declared signed centered DI in [-1, 1]. It does not change the score formula according to confirmation labels or outcomes.

At a permitted budget b, the prespecified score is:

```
S_b = (1 / b) * sum_{j=1..b} (d_j / sigma_j)^2.
```

Only already acquired observations enter this sum. The gate receives a bounded tuple/list copied at that checkpoint; it has no bank, waveform, truth label, damage-location parameter or future-query function. It refuses generators that could fetch unseen observations while being consumed. A finite positive scale is a numerical/reference requirement, not evidence that the residual is Gaussian; the subsequent rank comparison does not assume Gaussian residuals.

For complete healthy trajectory i, the calibration value is:

`M_i = max_{b in declared_checkpoint_prefix} S_{i,b}.`

At test prefix b, the conservative empirical reference rank is:

`p_b = (1 + count_i(M_i >= S_b)) / (N + 1).`

Ties use >=. Alpha is fixed at 0.05. With exactly 19 blocks, the minimum attainable p is 0.05: an alert requires the prefix score to exceed **every** calibration maximum. A tie with the largest calibration value gives at least 0.10 and cannot alert. This coarse resolution must be shown with every result. There is no interpolation of p-values to manufacture stronger evidence.

What the maximum protects, and what it does not

For a hypothetical complete test trajectory, its maximum `M_test` dominates each of its permitted prefix scores. Because the reference-rank p-value falls as the score increases, an alarm at any checkpoint implies an alarm for the complete test maximum. Thus calibrating complete maxima addresses repeated looks at these fixed checkpoints without treating the individual path scores as independent. Runtime scoring still needs only the acquired prefix.

The familiar finite-sample conformal rank argument would require the complete healthy calibration trajectories and a new healthy trajectory to be exchangeable, with the same frozen fitting procedure, controller, horizon, score and condition distribution. Model fitting must be separate from rank calibration. Adaptiveness within a trajectory is compatible with treating the complete trajectory as the calibration unit when the same closed-loop procedure generates each exchangeable unit.

The proposed cooling calibration versus warming confirmation split has **unverified exchangeability**. Thermal hysteresis, ordering, sensor coupling, and dependence among temperatures on one plate can violate it. Accordingly, the experiment reports an **empirical reference rank** and observed outcomes, not guaranteed 95% field coverage, a certified 5% false-alarm rate, or a damage posterior. The result cannot certify temperatures outside the qualified reference range. It also does not protect a search across different controllers, folds or tuned score variants, nor an indefinitely extended acquisition horizon, unless a separate suitable calibration protocol is declared.

Three states and safeguards

The gate returns the following explicit states:

State	Required interpretation
`supported_anomaly`	Valid selected observations exceed the empirical healthy trajectory reference at the frozen alpha; further verification is required.
`background_consistent`	The acquired prefix is not exceptional by this reference rank; it does not prove the plate is healthy or authorize a healthy release.
`unsupported`	The reference, model quality, numeric data, provenance, checkpoint or sample resolution cannot support this comparison.

The caller must qualify the transported reference and model quality. A failed acquisition remains an opportunity in the trace and makes the comparison unsupported; it cannot be removed from the prefix to strengthen an alert. Empty prefixes, unexpected checkpoints, NaN/Inf, nonpositive scales, repeated paths, unsupported accepted flags and numeric overflow all return unsupported.

Calibration binds lowercase SHA256 identities for the model, controller, protocol and fitting-reference library, plus the exact declared checkpoint prefix. A change requires a matching new calibration. Complete blocks have unique bounded IDs and unique raw-member SHA256 values. Testing a block ID or raw artifact already used for calibration is unsupported. These checks detect accidental reuse; hashes do not establish authenticity or specimen independence. The calibration snapshots caller lists and stores immutable tuples, so later edits to a trace or exported JSON cannot change its rank distribution.

An anomaly decision never supplies localization support. A qualified spatial response model, query accounting, a fresh geometrically distinct verification measurement, and the apparatus acquisition contract remain separate conditions for a supported location. The gate does not replace the direct-path calibration/latch checks on the ESP32 device or claim that 40 kHz ultrasonic data validate a 16 kHz audio-range probe.

Minimal experimental API

```
binding = GateBinding(model_hash, controller_hash, protocol_hash, reference_hash)
rows = tuple(SelectedObservation(path, corrected_di, healthy_sigma, accepted)
             for path, corrected_di, healthy_sigma, accepted in acquired_trace)
healthy = HealthyTrajectory(block_id, raw_member_sha256, rows, binding)
calibration = calibrate_reference_gate(healthy_trajectories, binding, alpha=.05)
decision = evaluate_reference_prefix(
    calibration, rows[:budget], binding,
    block_id=test_block_id, source_sha256=test_raw_sha256,
    reference_qualified=reference_ok, model_quality_accepted=model_ok,
)
```

`healthy_trajectories` must contain complete healthy replays reaching the binding's declared horizon (16 in the healthy-feasible experiment, 32 by default). The caller records all failures and the actual number of valid blocks. It derives hashes from the frozen implementation/configuration and exact source members. A hand-written hash string is not a validation of the fitted model.

Required reporting and tests

Report all 19 planned temperatures, acquired/source hashes, unsupported banks, actual calibration N, minimum p, selected paths, rejected opportunities, every checkpoint score/rank/state, earliest anomaly budget and separate localization outcomes. Retain negative or inconclusive comparisons and all source/code identities. Cooling/warming direction and one-specimen dependence are explicit limitations. No confirmation-dependent score or threshold substitution is allowed in the confirmatory table; any later variant is exploratory.

Focused tests in `backend/tests/test_trajectory_reference_gate.py` cover block order invariance, exact rank resolution, conservative ties, early-versus-final maxima, maximum domination of every checkpoint, causal prefix behavior, generator refusal, signed score equivalence, invalid data, rejected/repeated paths, identity reuse, changed bindings, insufficient N, complete-calibration requirements and immutable snapshots. They validate implementation behavior, not the physical exchangeability assumption or detection sensitivity. The feasibility revision adds complete 16-path calibration, unchanged rank resolution, invalid checkpoint-prefix rejection, and mismatch/extended-look rejection between 16- and 32-path bindings.

Existing mathematics and limited invention discussion

Conformal ranking is established mathematics. Shafer and Vovk's 2007 tutorial explains its assumptions and construction. Agia et al., *Unpacking Failure Modes of Generative Policies: Runtime Monitoring of Consistency and Progress* (CoRL 2024), explicitly calibrate whole closed-loop trajectories to handle correlated sequential states/actions; see its primary paper and authors' repository. The earlier 2011 trajectory anomaly paper also applies sequential conformal detection to incomplete trajectories. The gate cannot plausibly claim invention of conformal p-values, maxima or generic sequential anomaly detection.

A technical interaction to investigate in a narrow apparatus claim is the ordered combination of: (1) provenance-bound reference capture with source timing qualification; (2) healthy-only path/condition transport and uncertainty that can redirect an acquisition toward reference repair; (3) one-at-a-time feasible actuator/receiver selection under a fixed opportunity budget; (4) calibration of the actual resulting healthy controller trajectories; (5) a three-state gate that prevents an unsupported capture from advancing a structural claim; and (6) a fresh diverse verification capture bound to the same record. These are **candidate technical relationships**, not a novelty opinion, a proposed patent claim known to be allowable, or evidence of an inventive step. The interaction needs closer prior-art searching, actual hardware execution and comparative results before it can strengthen a grant assessment.